

Two-Stage Vector Sketch Generation Using Structure-to-Detail Optimization

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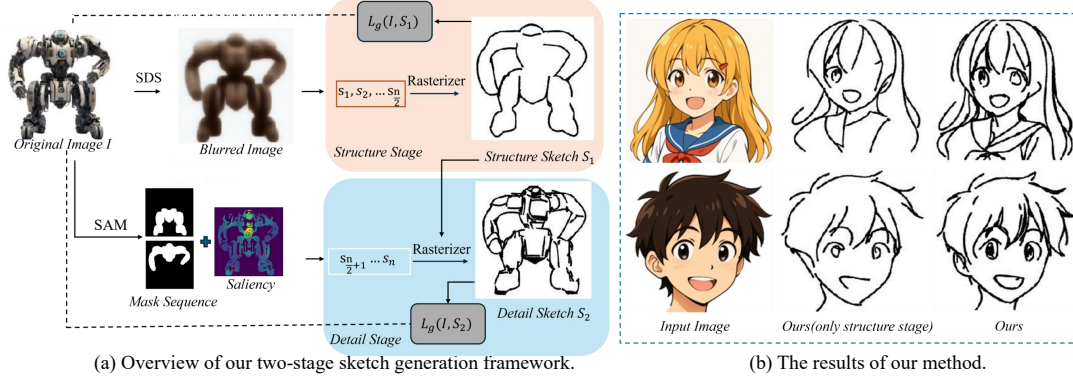


Figure 1: (a) Workflow of our proposed method. In the structure stage, the model initializes strokes to capture the global structure. In the detail stage, additional strokes are introduced to enrich local details. (b) Our method progressively generates vector sketches by first capturing the global structure and then refining local details.

ACM Reference Format:

Yuan Ma and Haoran Xie. 2025. Two-Stage Vector Sketch Generation Using Structure-to-Detail Optimization. In *SIGGRAPH Asia 2025 Posters (SA Posters '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3757374.3771449>

1 Introduction

With the recent advancements in Vision-Language Models (VLMs), the sketch generation domain has witnessed a paradigm shift that reduces reliance on human-drawn datasets and moves toward leveraging the rich priors encoded in pretrained models [Vinker et al. 2022]. Although these models can achieve impressive artistic results, they often produce sketches that lack accurate structural representation and sufficient detail. In contrast, our observation of human artists reveals a different strategy: they typically begin by outlining the global structure and then progressively refine local details.

Motivated by these findings, we propose a two-stage optimization framework inspired by the progressive abstraction strategy observed in human drawing. In the structure stage, we use Score

Distillation Sampling (SDS) to obtain a blurred version of the input image, providing a global-structure prior to capture a coarse but semantically meaningful outline of the subject. In the detail stage, we switch to the original high-resolution image and optimize the remaining strokes to refine local details. Throughout two stages, the proposed method guides the optimization using a CLIP-based perceptual loss, which enforces semantic consistency between the generated sketch and the input image.

2 Method

In this work, we define a sketch as a set of n black strokes $\{s_1, \dots, s_n\}$ placed on a white background. Each stroke is represented as a 2D Bézier curve defined by four control points. An overview of our method is shown in Fig.1(a). Given an input image, our goal is to generate a corresponding sketch that preserves both the semantic meaning and structural integrity of the subject. To mimic the human sketching process, we adopt a two-stage optimization strategy from structure to detail. In the structure stage, we optimize the stroke parameters by minimizing the loss between the blurred image and the structure sketch S_1 , ensuring that the generated sketch captures accurate global structure. In the detail stage, we build upon the structure sketch by refining local regions, thereby enriching details and producing a visually appealing detail sketch S_2 . Specifically, at each step of the optimization, we employ a differentiable rasterizer to optimize the stroke parameters directly under the guidance of a CLIP-based loss.

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SA Posters '25, Hong Kong, Hong Kong

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ACM ISBN 979-8-4007-2134-2/25/12

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2.1 Loss Function

Inspired by CLIPasso [Vinker et al. 2022], we adopt the pretrained image encoder of CLIP as our perceptual supervision. Since CLIP was trained on a wide variety of image-text pairs, it is capable of encoding high-level semantic information from both natural images and abstract sketches. To measure the geometric similarity between the image and the sketch, and consequently allow control over the appearance of the output, we compute the L_2 distance between intermediate level activations of CLIP:

$$L_g(\mathcal{I}, \mathcal{S}_i) = \sum_l \|\text{CLIP}_l(\mathcal{I}) - \text{CLIP}_l(\mathcal{S}_i)\| \quad i = \{1, 2\} \quad (1)$$

where CLIP_l is the CLIP encoder activation at layer l . Specifically, we use layers 3 and 4 of the ResNet101 CLIP model.

2.2 Optimization Stage

2.2.1 Structure Stage: We adopt the SDS loss to simplify image appearance via gradient-based optimization. Given a text prompt y , SDS guides the image toward semantic alignment using a pretrained diffusion model. The gradient can be approximated as follows:

$$\nabla_{\theta} \mathcal{L}_{\text{SDS}}(\theta) \approx \mathbb{E}_{t, \epsilon, c} \left[\omega(t) \left(\epsilon_{\phi}(x_t, t, y) - \epsilon \right) \frac{\partial g(\theta, c)}{\partial \theta} \right] \quad (2)$$

where x_t is the noisy image at timestep t , ϵ is Gaussian noise, and ϵ_{ϕ} is the noise predicted by the diffusion model. Since t and ϵ vary at each step, the predicted noise ϵ_{ϕ} also fluctuates, causing the gradients to point in inconsistent directions. Averaging over these directions suppresses high-frequency details, leading to simplified, feature-smoothed images. The initial sampling points are obtained by multiplying the saliency map with the XDoG edge map of the input image. We then optimize the $n/2$ stroke parameters based on this blurred image.

2.2.2 Detail Stage: In the detail stage, to ensure that no fine details are lost, we first apply Segment Anything Model (SAM) to segment the original image. Then, based on the saliency strength within each mask, we allocate strokes and initialize their parameters accordingly. Finally, we sequentially optimize the stroke parameters mask by mask, while the structure strokes remain fixed.

3 Results

Figure 2 illustrates the visual differences between our method and CLIPasso across various input images. The first column shows the input images, and the second column displays the blurred images produced. The third and fourth columns present the results from the structure and detail stages of our method, respectively. The final column shows the sketches generated by CLIPasso [Vinker et al. 2022]. Compared to CLIPasso, our method generated sketches that more accurately preserve the semantic structure and salient features of the original image, resulting in outputs that are both faithful to the source and visually more appealing.

For evaluation, we employ two different classifiers to assess the synthesized sketches: a CLIP ViT-B/32 zero-shot classifier with text prompts of the form “A sketch of a(n) [class-name]”, and a ResNet34 classifier trained on the Sketchy database [Sangkloy et al. 2016]. Table 1 compares the sketch classifier recognition accuracy of our sketches to that of CLIPasso [Vinker et al. 2022] based upon 100

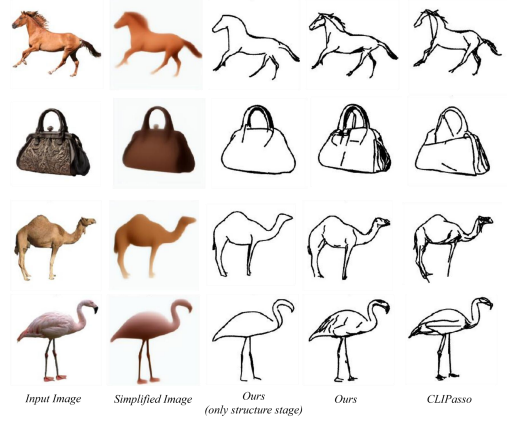


Figure 2: Qualitative comparison between the results generated by our method and CLIPasso [Vinker et al. 2022].

randomly selected images of 5 categories from the SketchyCOCO dataset [Gao et al. 2020]. Our method achieves higher accuracy than CLIPasso.

Table 1: The comparison results with CLIPasso.

	CLIPasso	Ours (only structure stage)	Ours
ViT-B/32	89%	85%	94%
ResNet34	63%	56%	71%

4 Conclusion

We proposed a two-stage vector sketch generation framework. The structure stage captures global composition with coarse strokes, while the detail stage leverages SAM segmentation and sequential mask-by-mask optimization to refine fine-grained details. Guided by a differentiable rasterizer and CLIP-based perceptual loss, our method can produce vector sketches with clearer structure and richer details.

Acknowledgments

This work is funded by New Energy and Industrial Technology Development Organization (NEDO) JPNP20017, and also supported by JST SPRING JPMJSP2102 and JST BOOST JPMJBY24D6.

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